

Machine Learning Pipeline for Loan Default Prediction

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CS611 Machine Learning Engineering
Assignment 2



Business Problem, Objective & Scope

Problem: Default is the biggest source of credit loss, it must be caught at the point of application. After approval, the loss is locked in.

Objective: Predict default probability at application.

Scope: Production end to end ML pipeline

12,500

Total Loans

28.9%

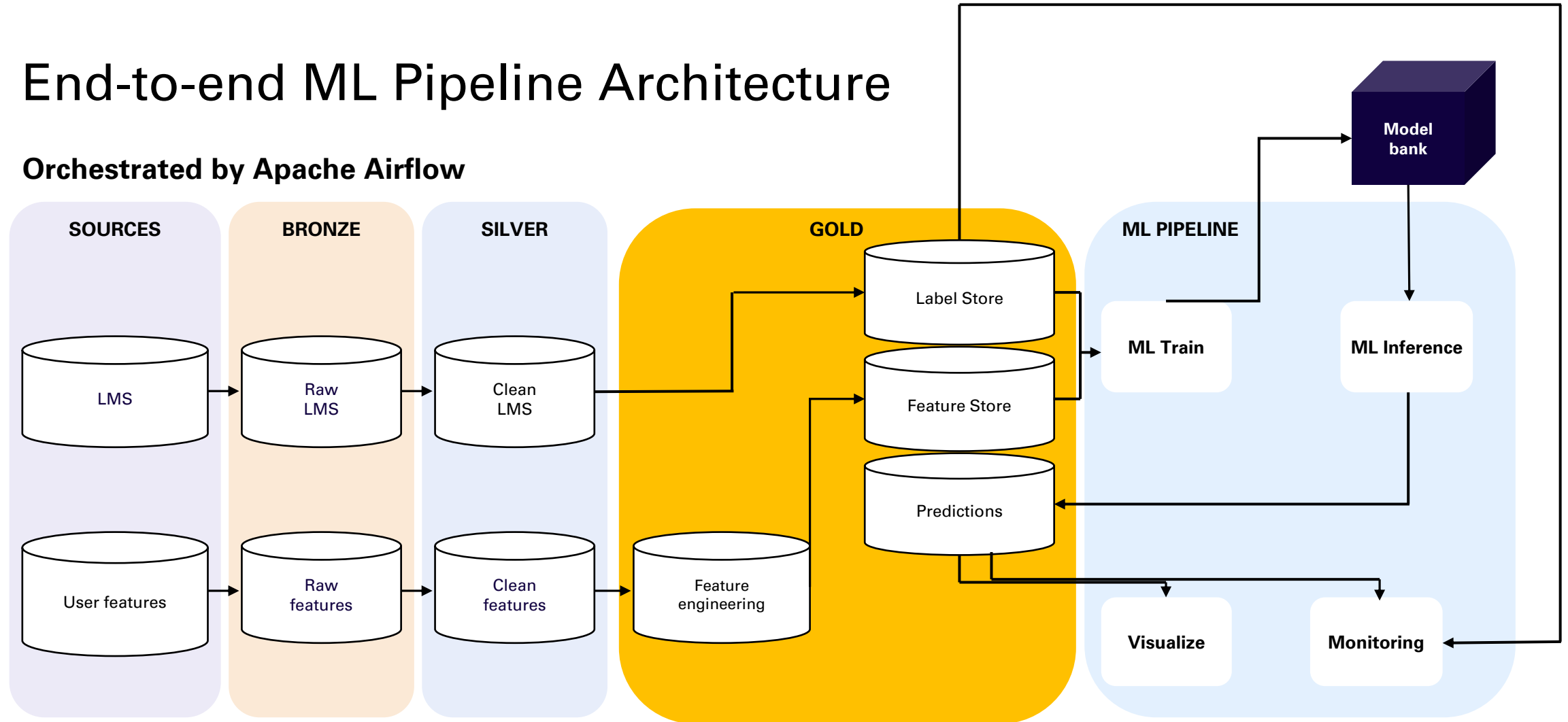
Observed Default Rate

4

Sources: LMS, attributes,
financials, clickstream

End-to-end ML Pipeline Architecture

Orchestrated by Apache Airflow



Dataset Split Design

- **Target leakage** - label comes from the LMS only, no outcome fields. (overdue / paid / balance)
- **Temporal leakage** - clickstream is aggregated strictly pre-loan-start, and the split is chronological by snapshot.

Chronological 70 / 10 / 10 / 10 split (by snapshot date) - OOT = most recent 10% (~2024-05 to 2024-06)

TRAIN 6,282	VAL 897	TEST 898	OOT 897
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Model Training & Selection

Training Set:

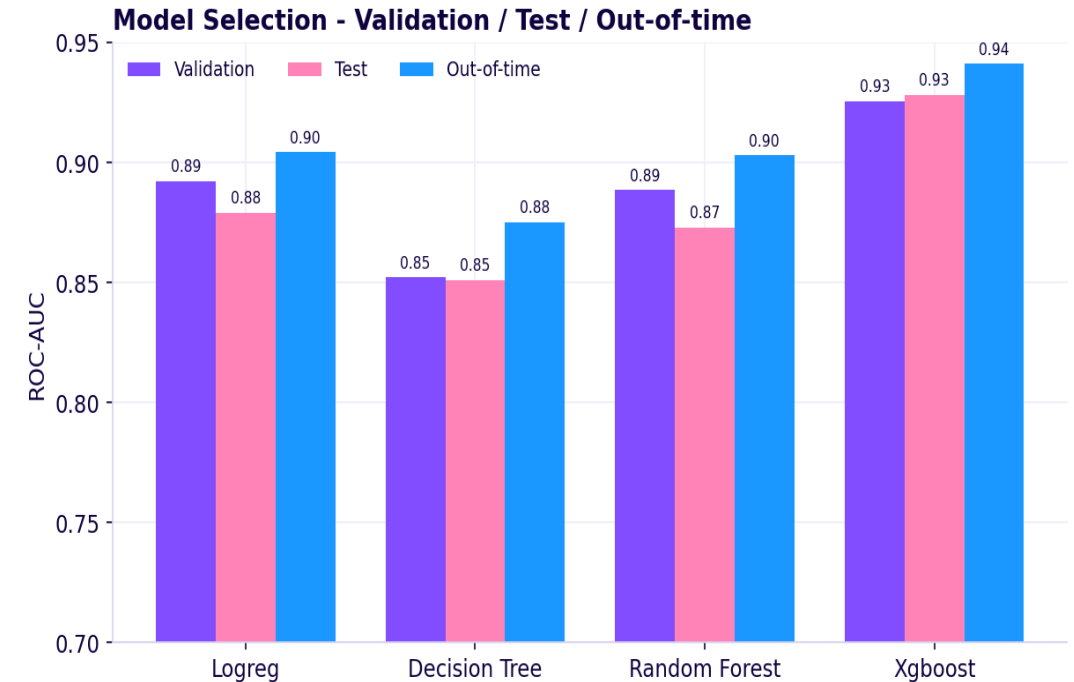
- 8,974 matured loans
- 52 features
- 28.9% default

Candidates:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. XGBoost

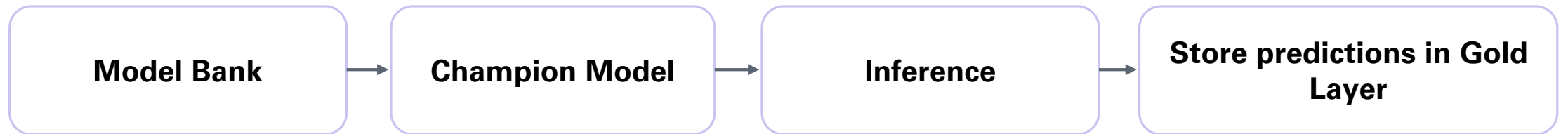
Preprocessing: Median-impute missing values, scaling numericals + one-hot encoding for categories

Selection: Best by validation AUC -> XGBoost (val 0.93, test 0.93, OOT 0.94).



Model bank. 4 artefacts per run + manifest.json (metrics, feature schema, best model).

Inference & Model Bank



Model Selection by Performance

Manifest names the best model + feature columns. Load by reference.



Versioned & Reproducible

Timestamped model-bank runs.



Schema-safe Scoring

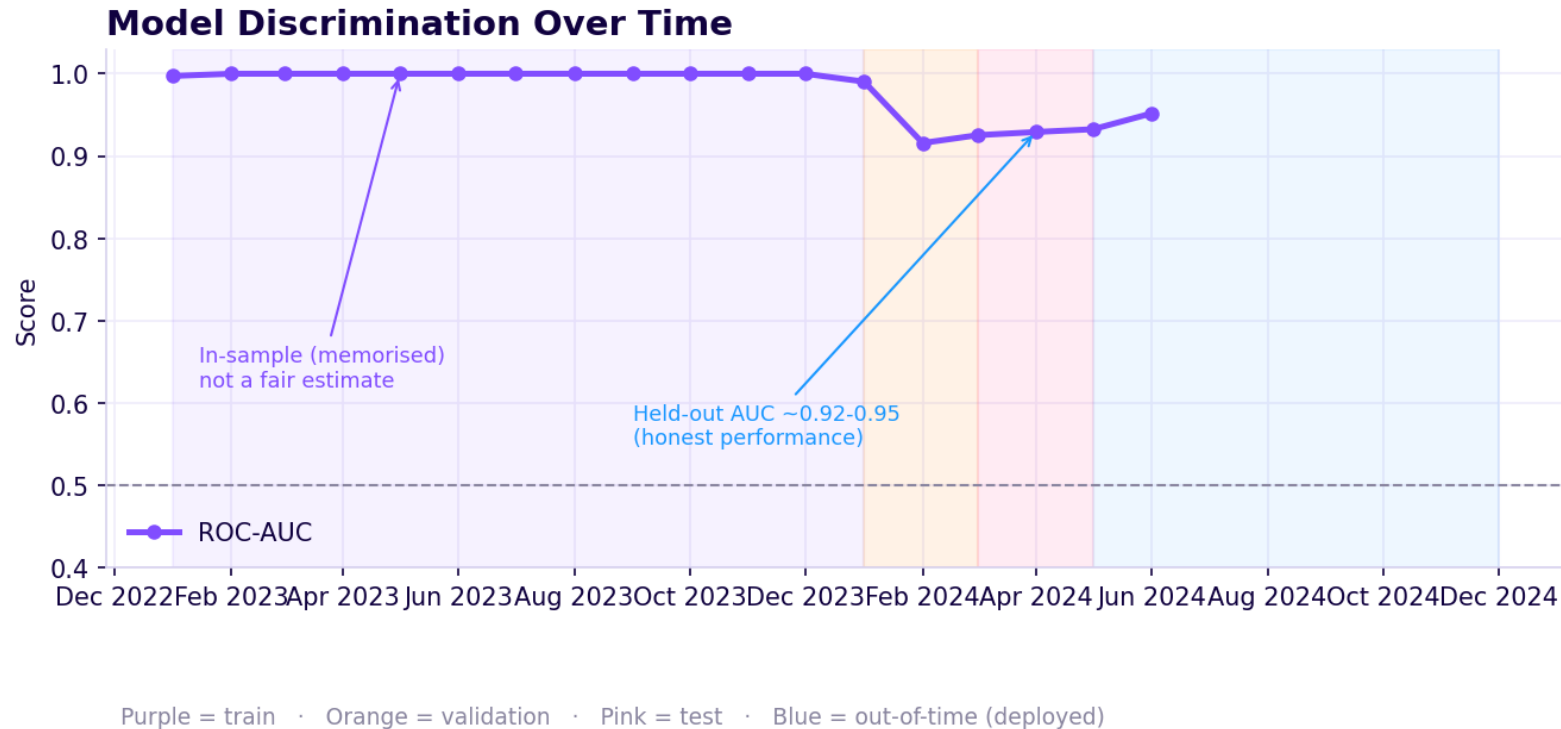
Rows reindexed per model schema.



Backfillable

Per-snapshot inference, backfill back-scores historical records

Monitoring - Model Performance Over Time



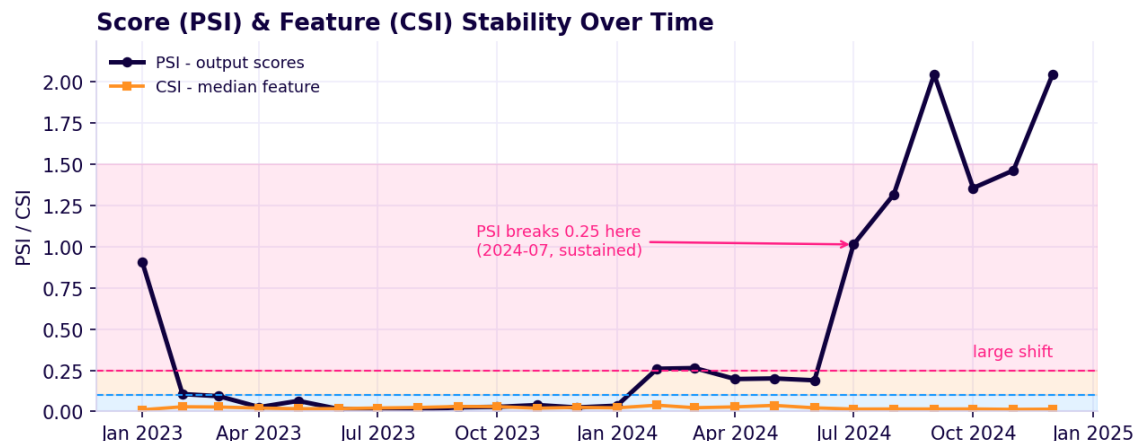
What we track:

- **ROC-AUC** - model evaluation (ranking power).
- **PSI** - stability of the output scores.
- **CSI** - stability of the input features.

Reading the chart:

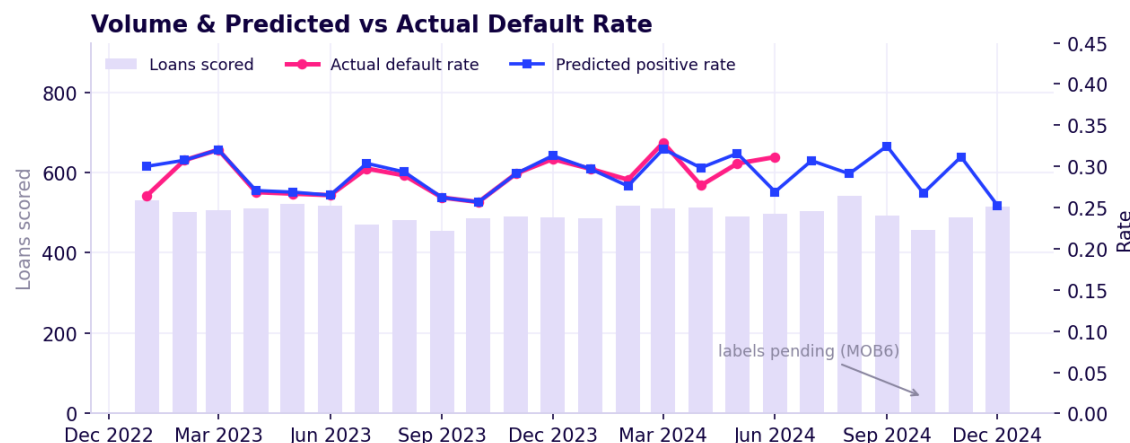
- Focus on OOT time-frame.
- AUC holds >0.90, good at generalizing.

Monitoring - Stability & Drift



PSI (Prediction) & CSI (Features)

Threshold	Risk Level	PSI	CSI
< 0.1	Low Risk	PSI breaks above 0.25 on July 2024	CSI remains stable below 0.05
0.1 to 0.25	Moderate		
> 0.25	High Risk		



Monitoring verdict

Since PSI shows that the prediction is drifting starting from July 2024, we should investigate and refresh on latest data to re-validate the model.

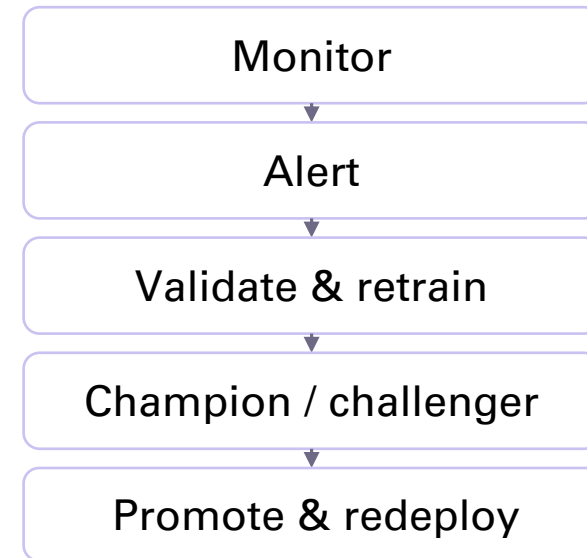
Model Governance, Refresh SOP

When and how to refresh the model in production

Refresh triggers

Stability	PSI > 0.25 sustained over 2 or more snapshots
Performance	Out-of-time AUC drops > 0.05 vs baseline
Scheduled	Quarterly retrain on latest matured data
Data change	Upstream schema / feature change

Refresh SOP



Technical Implementation & Summary



Docker

Portable, Reproducible Environment



Airflow
Orchestration, Scheduling & Backfill



PySpark

Scaling for Processing



Scikit-learn

Model Training & Selection



Parquet

Columnar Storage



GitHub

Version Control & Governance

Pipeline DAG (one run per monthly snapshot)



Key takeaways

Medallion Architecture · Reproducible (versioned model bank) · Monitored (performance + stability) · Governed (triggers + SOP)
Airflow Orchestration